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TO-Master: an LLM-agent framework for automated topology optimization

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Abstract

Topology optimization (TO) has become a mature computational design method, but using it still requires substantial manual effort in geometry preparation, mesh generation, boundary-condition assignment, solver setup, and postprocessing. This implementation barrier limits the use of TO outside expert workflows, even when differentiable finite element solvers are available. This work introduces **TO-Master**, a large language model (LLM) agent framework that turns finite-element-based TO into a conversational, tool-orchestrated workflow. From natural language instructions and optional mesh, geometry, or image inputs, the agent selects computational tools, constructs finite element TO models, checks meshes and boundary conditions, and launches sensitivity-based optimization with typed solver arguments. The framework supports generated and uploaded meshes, image-to-mesh conversion, 2D and 3D structural compliance minimization, thermal conduction, multiple load cases, stress-constrained optimization, and engineering geometries. Numerical experiments show that TO-Master can reproduce standard benchmark results and solve more complex engineering examples while returning optimized results, field distributions, convergence histories, and interactive artifacts without user-written code. An instruction ablation study further shows that tool-usage rules, internal reasoning guidance, and few-shot examples are critical for robust formulation under ambiguous user input. By combining LLM-agent orchestration with deterministic finite element and optimization tools, TO-Master removes the burden of trivial setup and routine model construction, lowers the modeling barrier of TO, and preserves a reliable numerical workflow. The TO-Master platform is available online at https://www.bohrium.com/en/apps/to-master.

Keywords: Topology optimization, LLM agent, Natural-language-driven modeling, Automated model generation

1 Introduction

Topology optimization (TO) is a computational design method that optimizes the material distribution in a prescribed design domain subject to boundary conditions, constraints, and objective functions. Classical methods include the homogenization-based method [1], the solid isotropic material with penalization (SIMP) method [2–4], the level-set method [5, 6], and the bi-directional evolutionary structural optimization method [7, 8]. These methods have been widely used for structural design, thermal design, and multi-physics inverse design problems. However, setting up a TO problem remains a nontrivial task. Users must specify

geometries, meshes, boundary conditions, material models, objectives, constraints, optimization parameters, and post-processing procedures. This process usually requires substantial problem-specific scripting and repeated manual modification, which limits the accessibility of TO workflows.

Recent advances in differentiable programming and automatic differentiation provide new opportunities for computational mechanics and inverse design. Automatic-differentiation-based TO frameworks such as `AuTO` [9] or differentiable finite element method (FEM) frameworks such as `JAX-FEM` [10] shows that finite element assembly, solution procedures, and objective evaluations can be embedded in automatic-differentiation-compatible computational graphs, making it possible to obtain sensitivities through automatic differentiation. These developments reduce the implementation burden of sensitivity analysis. Nevertheless, differentiable solvers still require users to formulate the computational problem correctly. In practice, the main difficulty often lies not only in solving the optimization problem constrained by partial differential equations (PDEs), but also in translating an engineering design intent into a complete and valid solver input.

Large language models (LLMs) have shown strong capabilities in language understanding, code generation, tool use, and multi-step reasoning. Chain-of-thought prompting improves reasoning over intermediate steps [11], while agentic frameworks such as ReAct enable LLMs to interact with external tools during problem solving [12]. Tool-augmented language models further demonstrate that LLMs can call external application programming interfaces (APIs) and use computational tools as part of their reasoning process [13]. These capabilities make LLM agents promising interfaces for scientific computing workflows, where user intent must be mapped to a sequence of structured computational operations. In computer-aided engineering (CAE), LLM agents have been explored for LLM-empowered CAE frameworks [14], automated structural analysis [15], computational fluid dynamics (CFD) case construction and solver execution [16, 17], and text-driven computer-aided design (CAD) generation [18, 19]. These studies show that LLM agents can reduce manual intervention in simulation setup and workflow orchestration. Recent and concurrent studies have also connected LLM agents with topology optimization, including LLM-based controllers for adapting SIMP continuation parameters [20], AutoSiMP for mapping natural-language descriptions to SIMP topology optimization problems [21], TO-Agents for preference-guided topology optimization with multi-agent visual feedback [22], and large-model-assisted conceptual structural design realization [23].

These works demonstrate the potential of LLM agents in TO, while two key challenges remain. First, many existing studies focus on relatively simplified structural compliance benchmarks in academic settings, while more general problem settings, engineering geometries, and user-level problem specifications remain less explored. Second, because a valid TO workflow requires reliable construction of boundary conditions, passive domains, physical constraints, and solver inputs, errors in these steps can produce plausible-looking but physically incorrect optimized structures.

In this work, we present **TO-Master**, an LLM-agent framework for topology optimization. The main contributions of this work are reflected in three aspects: first, an LLM-agent-based workflow that translates user-level natural language specifications into finite-element TO models; second, a tool-orchestrated implementation that couples model construction, verification, optimization, and postprocessing through typed deterministic tools; and third, a systematic evaluation across benchmark, engineering, and prompt-ablation cases. TO-Master takes user-level natural language specifications as input, optionally supplemented by mesh or geometry files, and supports 3D model and mesh generation from 2D image inputs. It automatically formulates finite-element-based TO problems, visualizes and verifies meshes and boundary conditions through human-in-the-loop confirmation, performs sensitivity-based optimization with a differentiable FEM backend, and reports optimized structures, field distributions, and convergence histories. The framework supports 2D and 3D structural compliance minimization, thermal conduction, multiple load cases, and stress-constrained optimization for both regular design domains and arbitrary engineering geometries, including user-provided unstructured meshes and 3D geometries reconstructed